

RISC-V Single Cycle Architecture II

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Lecture 8

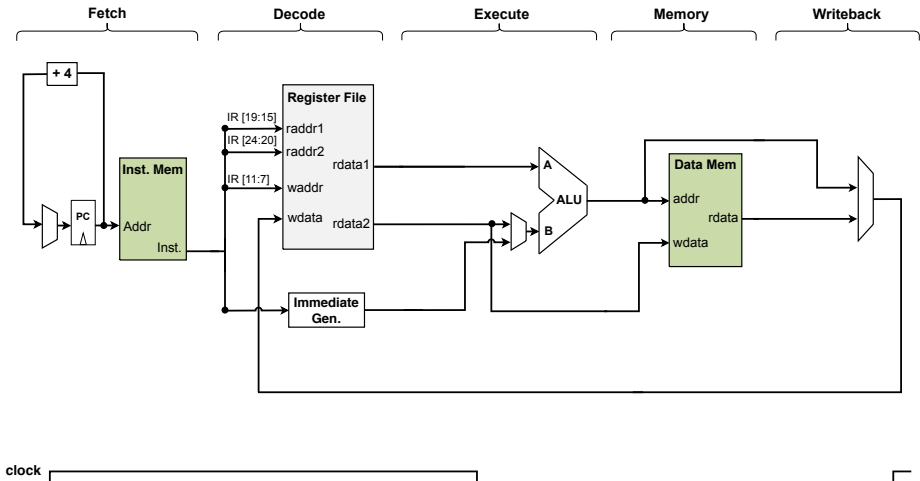
Electrical Engineering Department
University of Engineering and Technology Lahore

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- 2 B-type Datapath
- 3 U-type Datapath
- 4 J-type Datapath

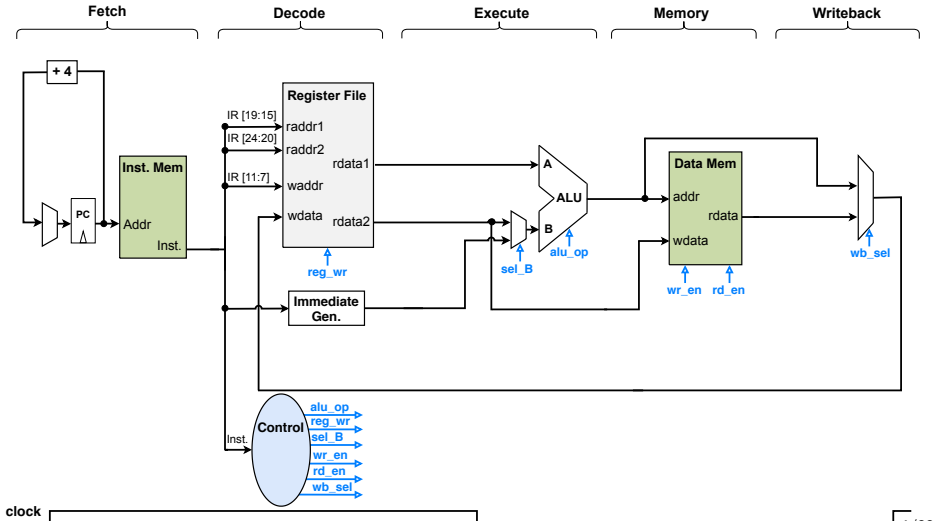
Review: ISA Supported So Far

- R, I and S type datapath

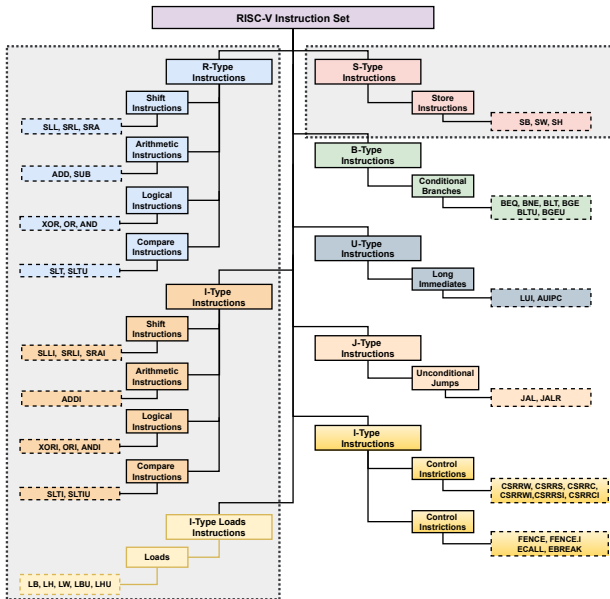


Review: ISA Supported So Far Cont'd

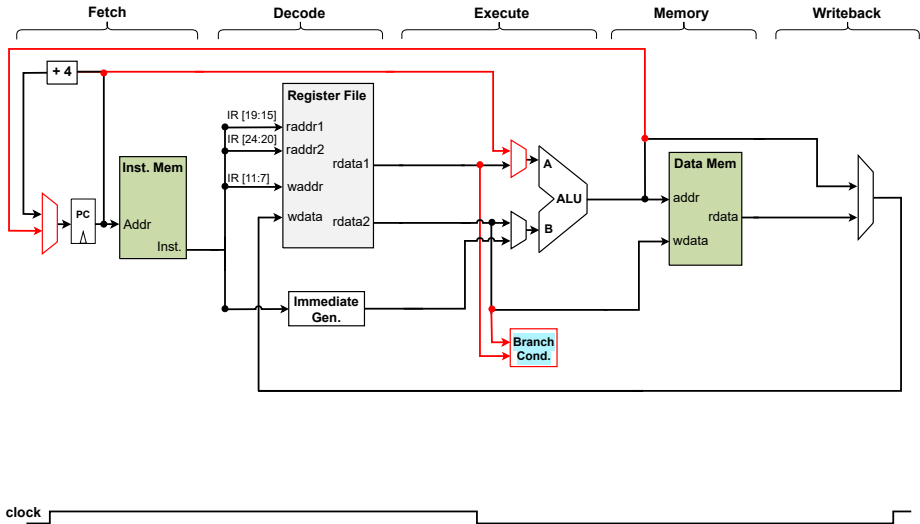
- R, I and S type datapath and control



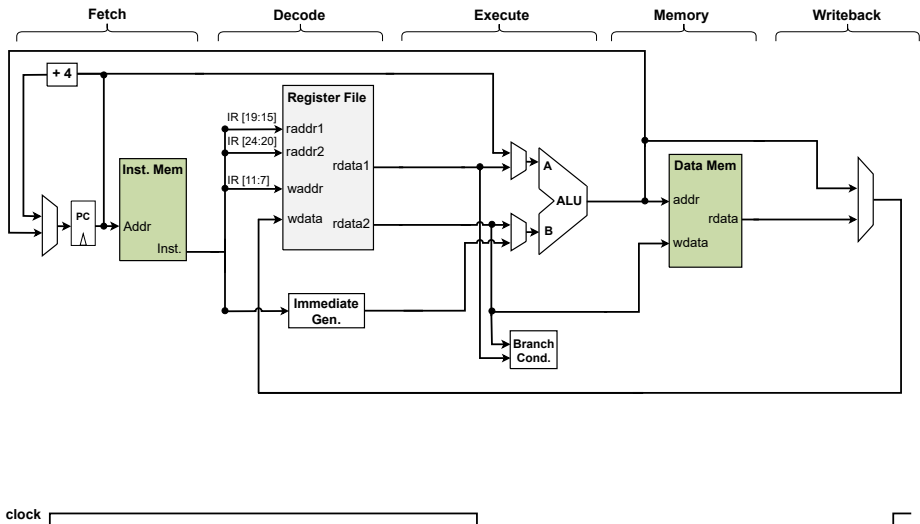
Review: ISA Supported So Far Cont'd



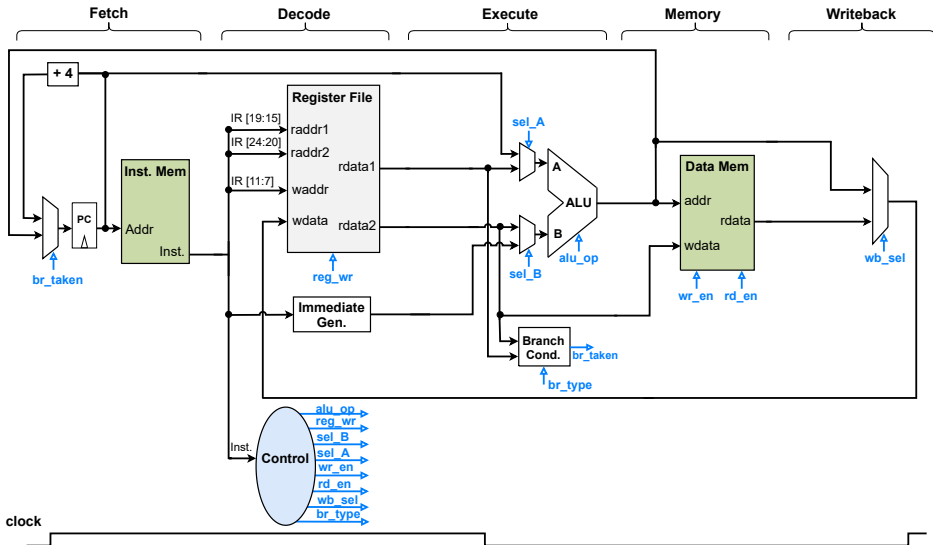
B-type: Conditional Branches



B-type: Conditional Branches Cont'd

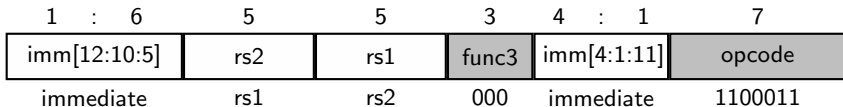


B-type: Datapath and Control



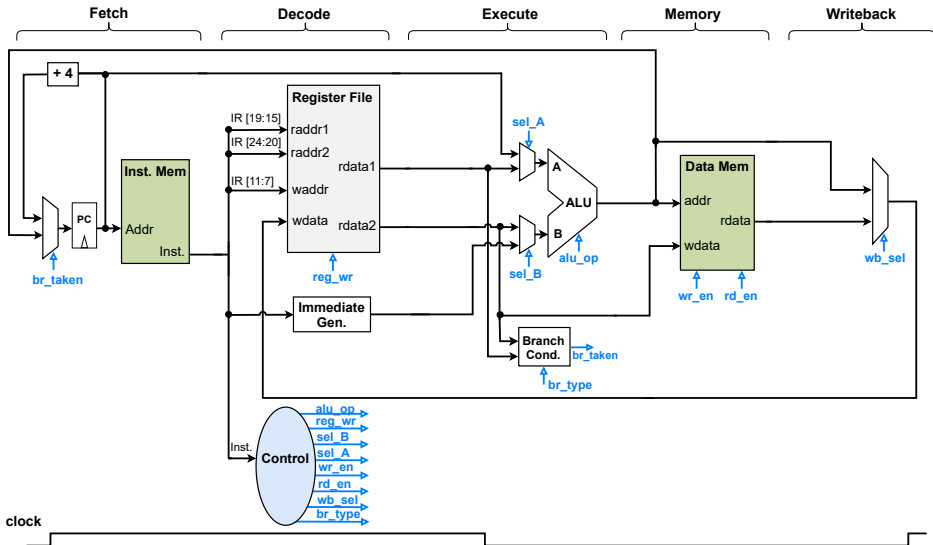
B-type: Branch Instruction Example

- B-type: Branch equal (BEQ) instruction

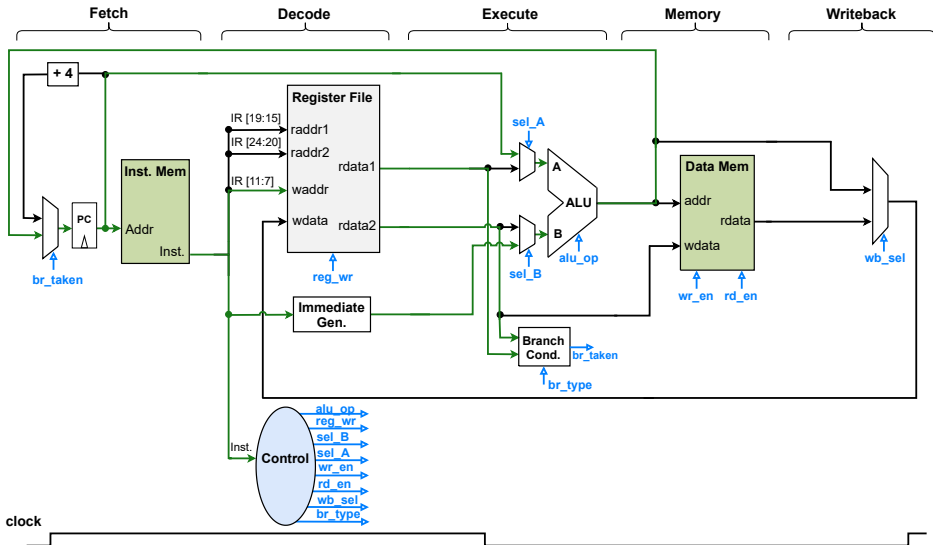


- $B-imm = \text{signExtend}(\text{inst}[31], \text{inst}[7], \text{inst}[30:25], \text{inst}[11:8], 1'b0)$
- $opcode = \text{BRANCH:}$
 $pc \leftarrow \text{compare}(\text{funct3}, rs1, rs2) ? pc + B-imm : pc + 4$
- $funct3 = \text{BEQ/BNE/BLT/BLTU/BGE/BGEU}$

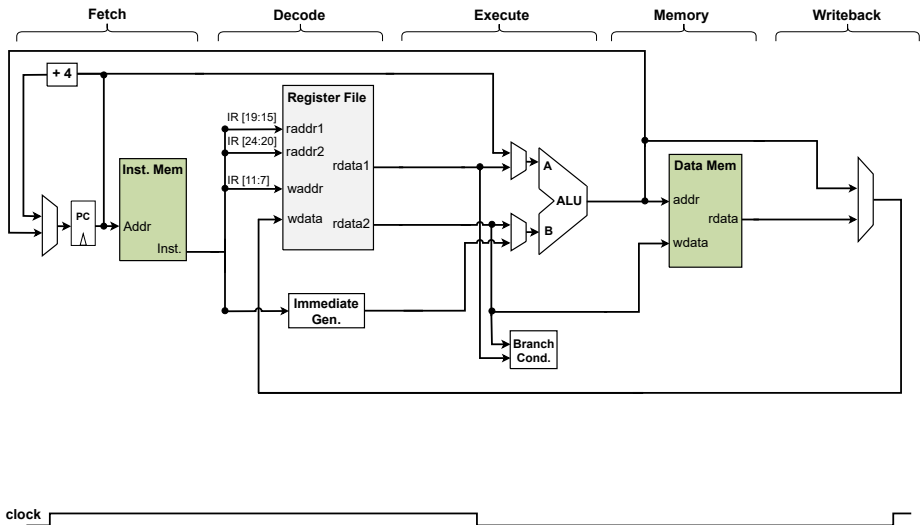
B-type: How Branch Equal Instruction Execute



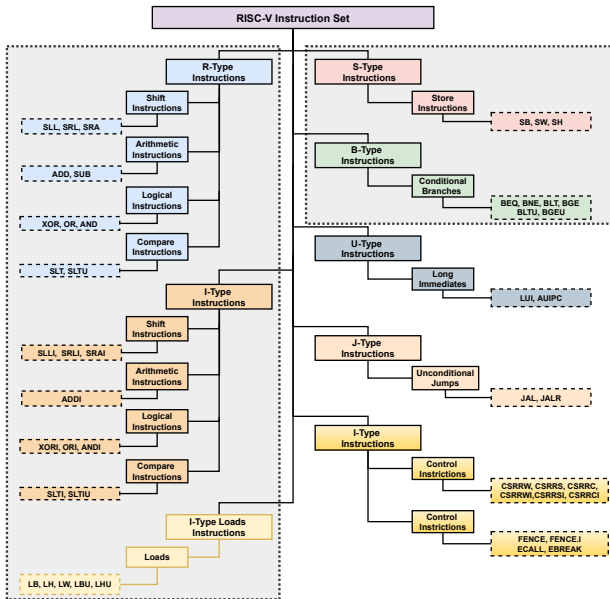
B-type: Branch Equal Instruction Execution



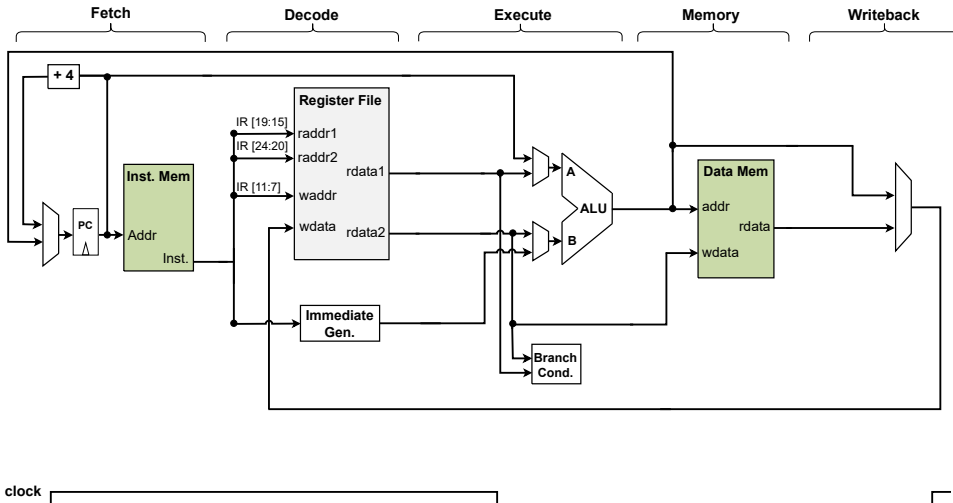
Datapath for R-, I-, S- and B-type



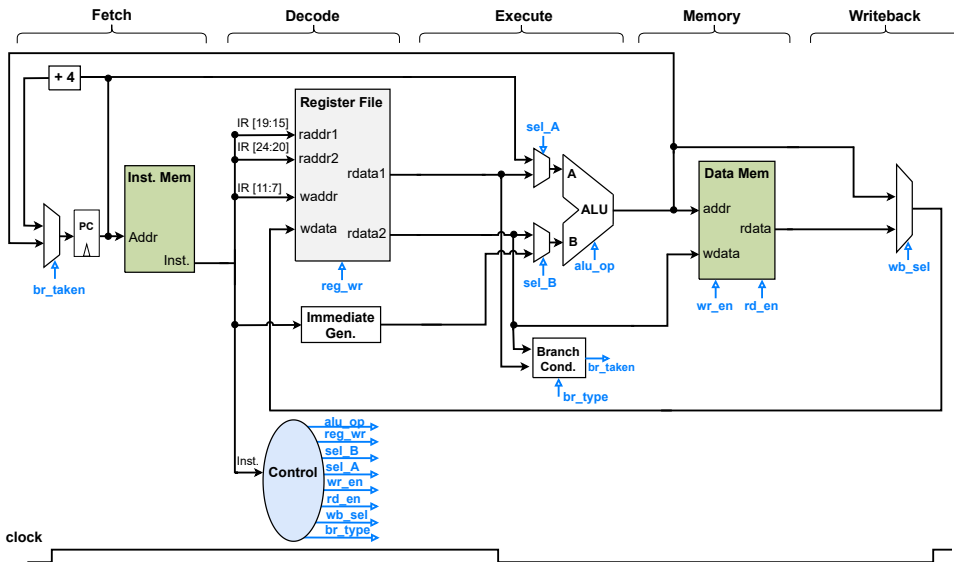
RISC-V ISA Supported by R-, I-, S- and B-type



U-type: Immediate Generation

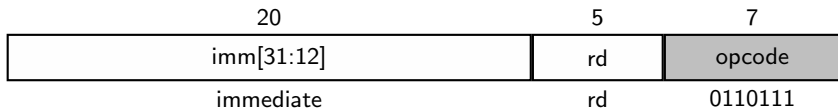


U-type: Datapath and Control



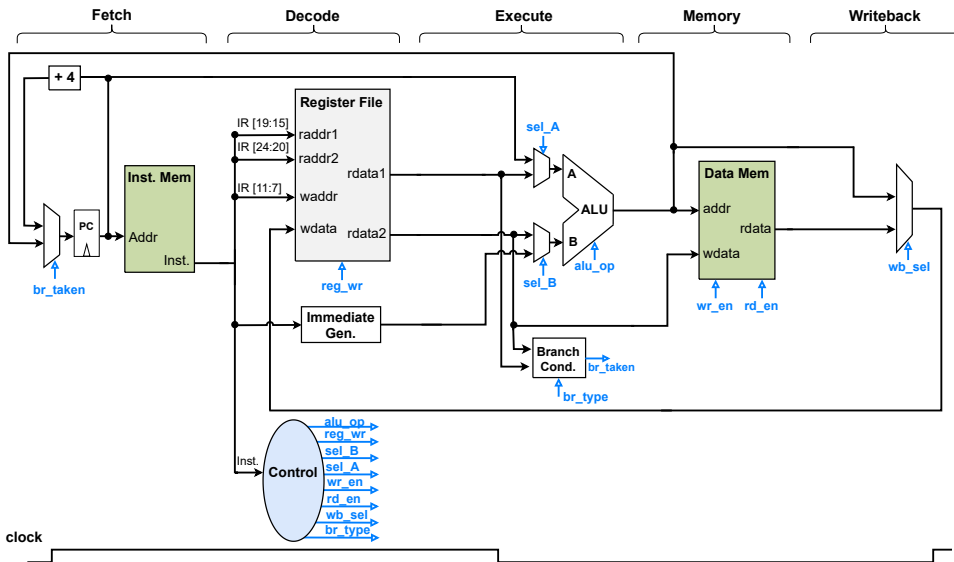
U-type: Load Upper Immediate Example

- U-type: LUI example

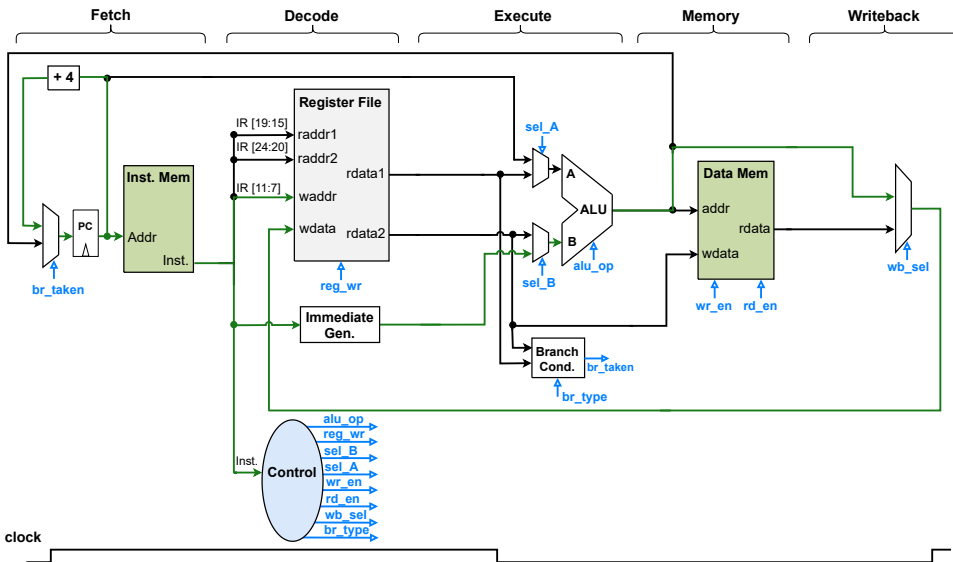


- $\text{U-imm} = \text{inst}[31:12], 12'b0$
- $\text{opcode} = \text{LUI} : \text{rd} \leftarrow \text{U-imm}$
- $\text{opcode} = \text{AUIPC} : \text{rd} \leftarrow \text{pc} + \text{U-imm}$

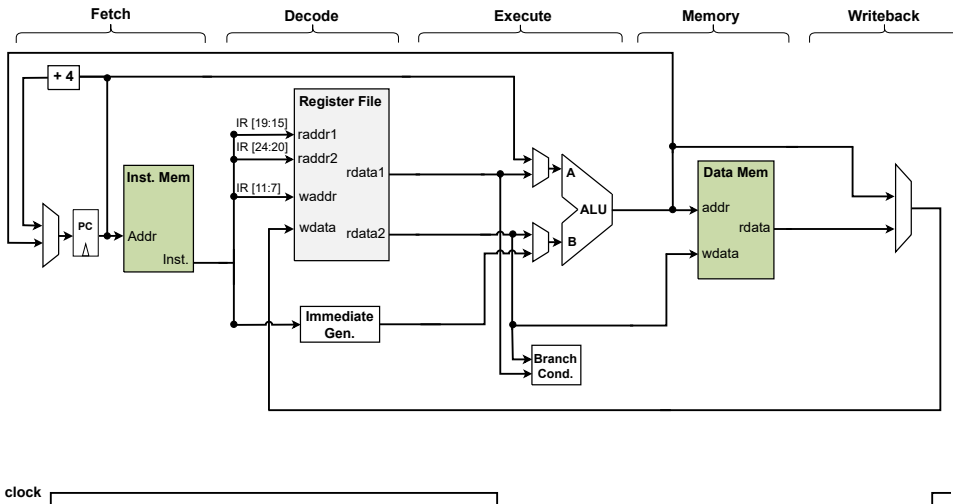
U-type: How LUI Instruction Execute



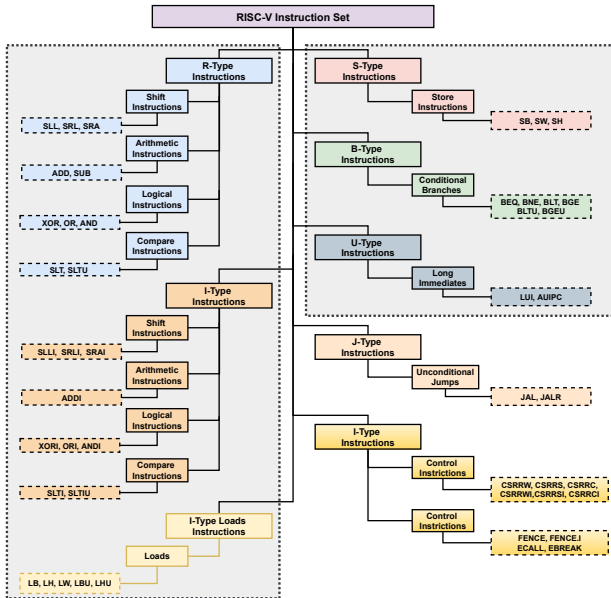
U-type: LUI Instruction Execution



Datapath for R-, I-, S-, B- and U-type

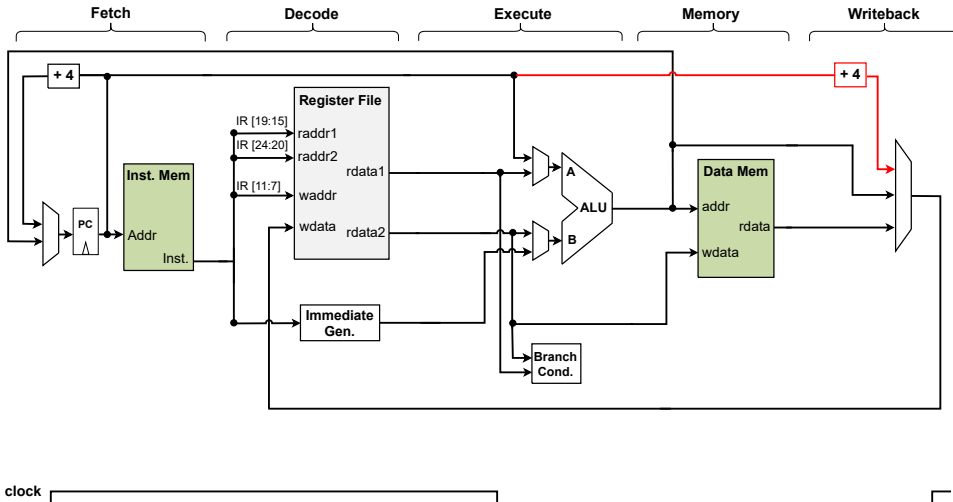


RISC-V ISA Supported by R-, I-, S-, B- and U-type

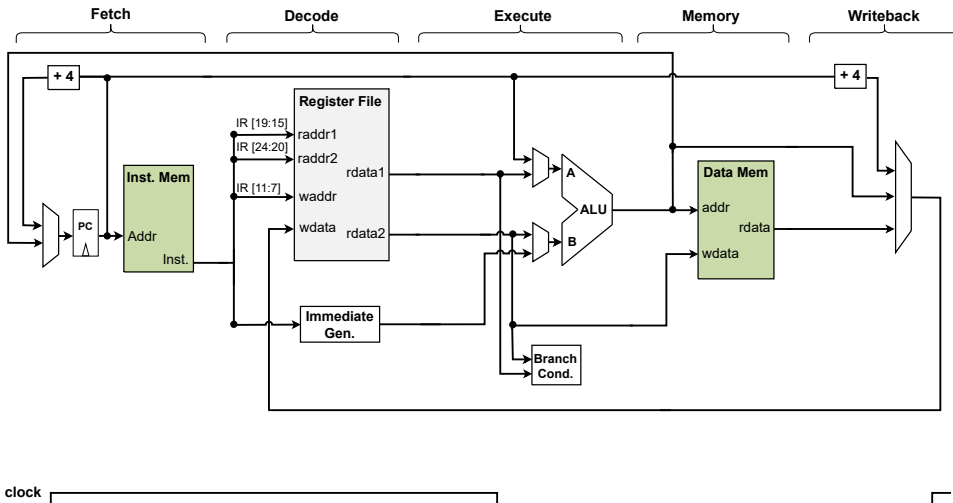


J-type: Jump Instructions

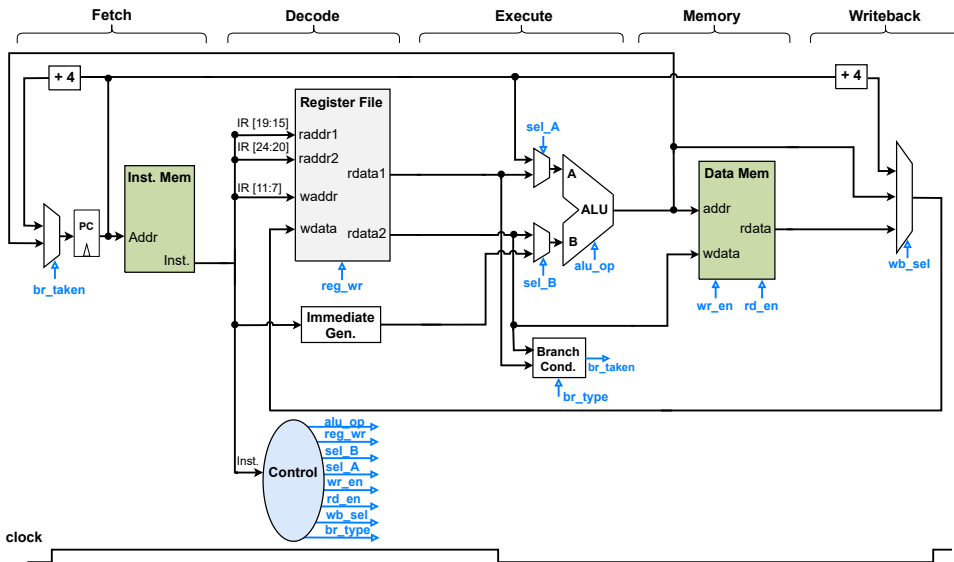
- Unconditional Jumps



J-type: Jump Instructions

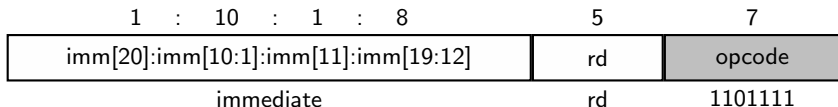


J-type: Datapath and Control



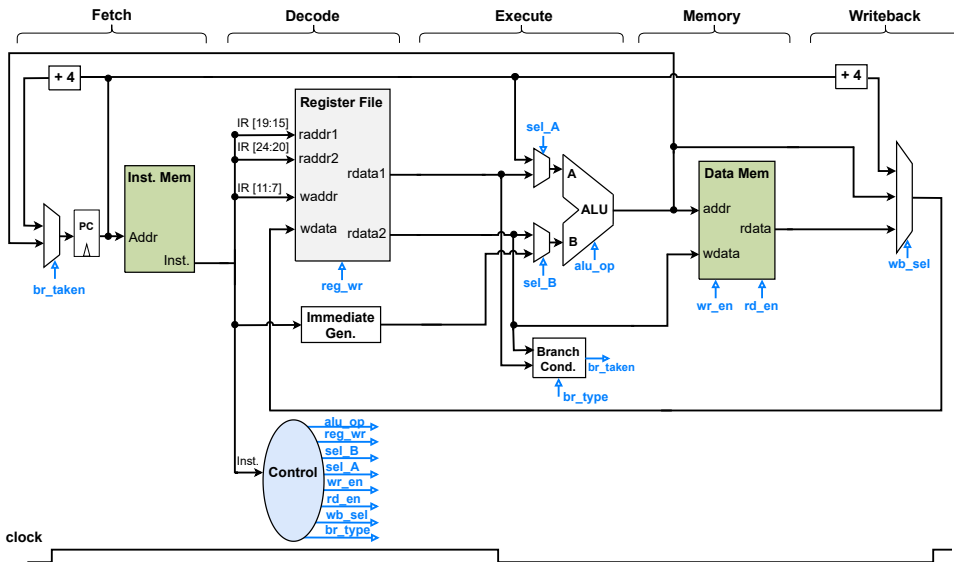
J-type: Jump and Link (JAL) Instruction

- Format: J-type

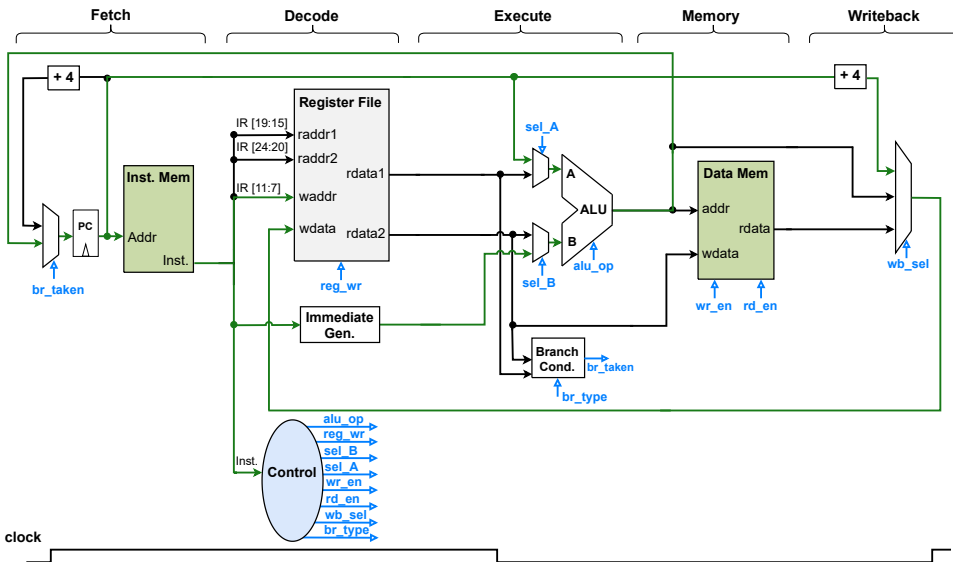


- $J\text{-imm} = \text{signExtend}(\text{inst}[31], \text{inst}[19:12], \text{inst}[20], \text{inst}[30:21], 1'b0)$
- $\text{opcode} = \text{JAL: } rd \leftarrow pc + 4;$
 $pc \leftarrow pc + J\text{-imm}$
- $\text{Jump} = \pm 1\text{MB range}$

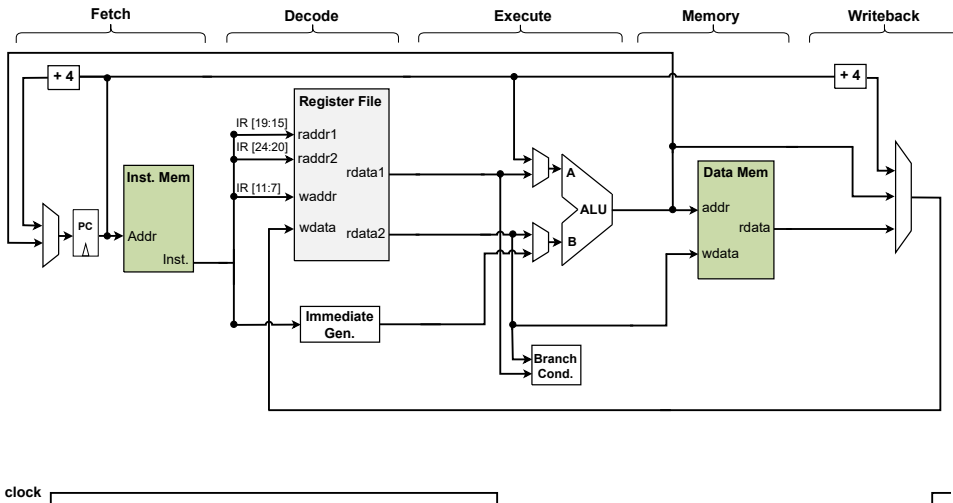
J-type: How JAL Instruction Executes



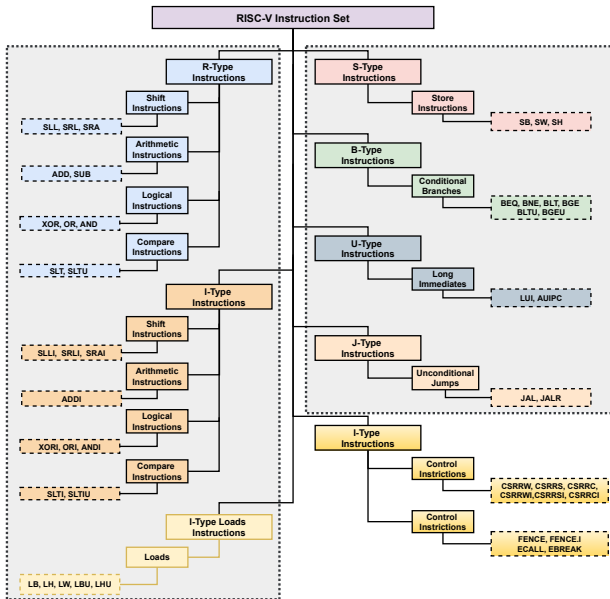
J-type: JAL Instruction Execution



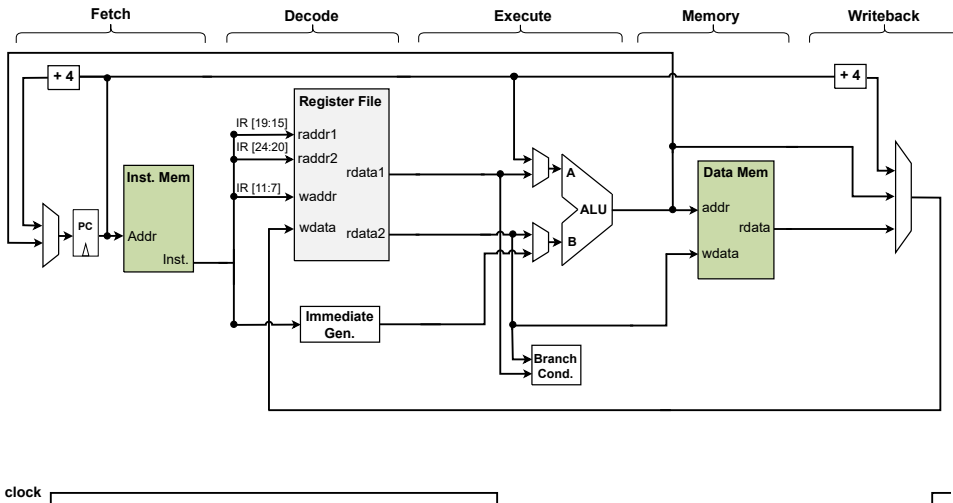
Datapath for R-, I-, S-, B-, U- and J-type



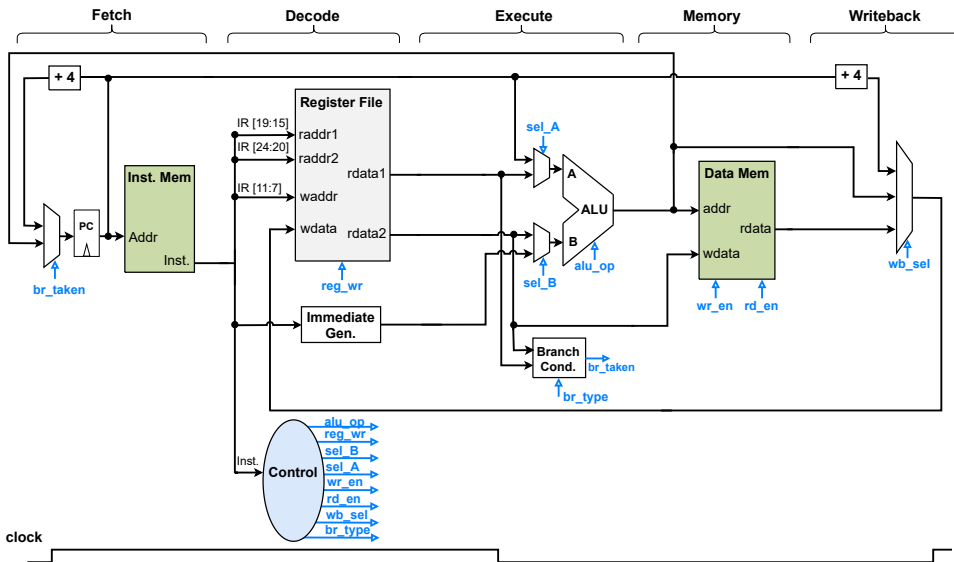
RISC-V ISA Supported by R-, I-, S-, B-, U- and J-type



Single Cycle Datapath



Single Cycle Datapath and Controller



Suggested Reading

- Read relevant sections of Chapters 2 and 4 of [\[Patterson and Hennessy, 2017\]](#).

Acknowledgment

- Preparation of this material was partly supported by Lampro Mellon Pakistan.

References



Patterson, D. and Hennessy, J. (2017).

Computer Organization and Design RISC-V Edition: The Hardware Software Interface.

Morgan Kaufmann.